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Possible Options for Unlocking and Securing U.S. Energy for AI Production

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About This Working Paper

This paper provides public comment in response to the February 6, 2025, request for information (RFI) by the National Science Foundation’s Networking and Information Technology Research and Development National Coordination Office on behalf of the Office of Science and Technology Policy (OSTP), concerning the development of an artificial intelligence (AI) action plan.¹ Our purpose is to share relevant expert knowledge on the subject areas prioritized in the RFI call. We focus on the “data centers,” “energy consumption and efficiency,” “regulation and governance,” and “innovation and competition” aspects, as described in the RFI, with particular emphasis on potential options to unlock energy for AI development. These options could improve energy capacity, which is key to accelerate AI progress and maintain U.S. competitiveness. Our target audience is OSTP. This paper is filed by Yusuf Mahmood, Christopher Byrd, Erika Somani, Konstantin Pilz, and Lennart Heim. The opinions presented in this paper are those of the authors and do not reflect those of RAND. This paper does not contain original research. We have not conducted a full analysis of the costs, benefits, or legal feasibility of the policy options described herein; we drafted the response based on our professional expertise and an assessment of published research.

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¹ National Science Foundation, “Request for Information on the Development of an Artificial Intelligence (AI) Action Plan,” *Federal Register*, Vol. 90, No. 24, February 6, 2025.

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Summary

Globally, AI data centers are facing a significant power shortfall, with an estimated fourfold increase in demand in 2025 compared to 2023.² If growth in AI chip demand continues at the current rate, this need could rise to 68 gigawatts (GW) by 2027, nearly doubling global data center power needs from 2022.³ AI chip production could increase by 1.3 - 2x annually through 2030, driving a projected demand of 327 GW by 2030.⁴ Per U.S. policy, the U.S. may need to unlock an additional 50 gigawatts of energy capacity in order to remain dominant in AI infrastructure by 2027.⁵ Yet, it is uncertain whether the U.S. is expanding its energy capacity quickly enough to meet growing demand from U.S. AI data centers.⁶ We refer to the resulting shortfall of energy supply in the U.S. as the “AI energy gap.”

The last administration took some steps to reduce the AI energy gap by issuing Executive Order (EO) 14141 in its final weeks. The Trump administration took additional steps to unlock domestic energy production in EO 14154: Unleashing American Energy.⁷ This working paper explores potential options to build on these executive actions across the following lines of effort:

- **A. Streamlining Permitting:** Reducing the regulatory burden for environmental reviews—especially for geothermal, nuclear, and traditional energy sources—to speed up the development of AI infrastructure.
- **B. Transmission Development:** Expediting the construction of necessary transmission lines to connect AI data centers to the energy grid, including leveraging DOE authorities to avoid state-level delays.
- **C. Federal Lands for AI Infrastructure:** Expanding the use of federal lands for AI infrastructure by expediting federal site selection processes and removing restrictions that mandate clean power procurement for AI data centers on leased federal land.
- **D. National Security Considerations:** Enhancing security measures for AI infrastructure developers, including the reporting of personnel risk factors and ensuring cybersecurity standards are met. Such measures help protect AI systems critical to national security.

² Dylan Patel, and Gerald Wong, “GPT-4 Architecture, Infrastructure, Training Dataset, Costs, Vision, MoE,” webpage, July 10, 2023.

³ Konstantin F Pilz, Yusuf Mahmood, and Lennart Heim, *AI’s Power Requirements Under Exponential Growth: Extrapolating AI Data Center Power Demand and Assessing Its Potential Impact on U.S. Competitiveness*, RAND Corporation, RR-A3572-1, 2025.

⁴ Jaime Sevilla and Edu Roldán, *Training Compute of Frontier AI Models Grows by 4–5x per Year*, Epoch AI, May 28, 2024.

⁵ Pilz, 2025.

⁶ Pilz, 2025.

⁷ Executive Order 14141, 2025.

Possible Options for Unlocking and Securing U.S. Energy for AI Production

The AI Energy Problem

The U.S. is at risk of losing its lead in data center and compute provider market share and is struggling to meet the growing energy demands to support AI training and inference needs.⁸ This could result in the U.S. losing its AI advantage and missing economic opportunities.⁹ In order to maintain and expand the current U.S. lead in AI, actions could facilitate the creation of new AI infrastructure, including new and larger AI data centers, additional facilities for energy generation/storage, and enhanced transmission capacity. This paper suggests potential options for OSTP to coordinate action across various agencies to expedite the development of this infrastructure. In formulating these options, we have drawn on our professional expertise and our assessment of published research. We did not conduct a full analysis of the costs, benefits, or legal feasibility of these options. We suggest further study prior to implementing any of these options.

Predicted Energy Shortfalls

Exponential growth in AI is placing significant strain on U.S. power grids, with the global demand for AI data centers projected to reach 68 GW by 2027.¹⁰ This is nearly equivalent to California's total power capacity (86 GW) in 2022.¹¹ U.S. companies are already struggling to secure enough grid capacity for data centers to meet AI demand, a challenge expected to intensify as AI training and deployment continue to grow.¹² If U.S. companies are unable to secure sufficient power, they may relocate their data centers abroad, which could undermine the country's AI leadership and carry national security implications. Furthermore, there is a significant economic growth opportunity for the U.S., as global AI data center demand is projected to reach 327 GW by 2030—far surpassing currently total global capacity of just 88 GW in 2022.¹³ This growing demand presents a critical area of opportunity for the U.S. to capitalize on its leadership in AI and related infrastructure. A subset of the authors studied expected increases in AI's power demands by 2027 and 2030 and found that the U.S. may fall short of the amount of energy it needs to maintain AI infrastructure dominance.¹⁴

⁸ Pilz, 2025; Sastry, Girish et al, "Computing Power and the Governance of Artificial Intelligence," arXiv:2402.08797, February 13, 2024.

⁹ Under EO 14141, "AI infrastructure" includes AI data centers, energy generation/storage, and transmission facilities.

¹⁰ If exponential growth in chip supply continues.

¹¹ Pilz, 2025.

¹² Sebastian Moss, "Meta's Mark Zuckerberg Says Energy Constraints Are Holding Back AI Data Center Buildout," webpage, April 19, 2024.

¹³ Pilz, 2025.

¹⁴ Pilz, 2025.

Impact of Federal, State, and Local Regulations

One of the key challenges in constructing data centers and power infrastructure is permitting processes, which can cause significant delays in development. For example, state and local land-use permits often take years to clear for large construction projects. Additionally, grid interconnection requests for new generation resources can take four to seven years for power hookup.¹⁵ Finally, transmission line projects often face complex multistate permitting processes and local opposition, further delaying power delivery to data centers.¹⁶ The difficulty of securing local and state permits is compounded by environmental regulations that make the use of available power sources challenging – limiting data center developers’ opportunities to procure power quickly, and pushing them to rely on more challenging-to-scale renewable options. National Environmental Policy Act (NEPA) environment impact statement (EIS) completion time was on average 4.5 years.¹⁷ A recent U.S. Department of Commerce (DOC) survey of manufacturers identified the Clean Air Act (CAA), Clean Water Act (CWA), and Endangered Species Act (ESA) as among the most significant regulatory barriers to new production, affecting industries’ ability to scale inputs such as AI chips.¹⁸ As a result, U.S. companies are exploring expansion into countries with more accessible power and quicker permitting processes as potentially their only opportunity to remain at the AI frontier.¹⁹ To address these challenges, we suggest potential options to free up energy for AI data centers.

Potential Energy Policy Options

The following policy options may help unlock and secure domestic energy production to support expanded AI data centers by 2027. We organize these policy options around four key features: (A) streamlining federal permitting processes; (B) grid access and building transmission lines; (C) designating federal lands for AI infrastructure; (D) and national security-related requirements to support AI infrastructure developers.

A. Streamlining Federal Permitting Processes

It may be possible to streamline federal permitting for AI infrastructure projects by simplifying requirements and providing compliance assistance with laws such as National Environmental Policy Act (NEPA), the Clean Air Act (CAA) and the Clean Water Act (CWA).²⁰ EO 14141 introduced some changes to these processes, and also places special focus on deregulating geothermal and nuclear energy.²¹ These policies could be expanded by: (i) ensuring timely completion of the U.S. Department of Defense’s (DOD) programmatic environmental review;

¹⁵ Josh Saul, “Data Centers Face Seven-Year Wait for Dominion Power Hookups,” *Bloomberg*, August 29, 2024.

¹⁶ Brian Potter, Arnab Datta, and Alec Stapp, *How to Stop Environmental Review from Harming the Environment*, Institute for Progress, September 13, 2022.

¹⁷ Council on Environmental Quality, *Environmental Impact Statement Timelines (2010-2018)*, June 12, 2020.

¹⁸ This is a compilation of RFI responses from manufacturers, which relates to energy but may not be entirely transferrable. U.S. Department of Commerce, *Streamlining Permitting and Reducing Regulatory Burdens for Domestic Manufacturing*, October 6, 2017.

¹⁹ Zaeem Shoaib, “Big Tech Moves More AI Spending Abroad,” *Wall Street Journal*, May 23, 2024.

²⁰ Sections 7 and 9 reduce NEPA review burden via categorical exclusions and programmatic environmental review; assist projects with CAA air-quality permit compliance; and orders speedy processing of CWA permits; among other examples.

²¹ Geothermal power holds promise for rapidly supplying several gigawatts of capacity to AI data centers.

(ii) exploring NEPA categorical exclusions for natural gas and oil; (iii) initiating interagency consultation processes required for Endangered Species Act (ESA) exemptions; (iv) activating national security exemptions for the Clean Air Act and Clean Water Act; and (v) considering Defense Production Act (DPA) authorities to avoid triggering federal and state regulations.

Option 1: Ensure timely completion of the DOD’s programmatic environmental review

Establishing federal sites for data centers obviates many state and local land use permits—and may speed data center development timelines significantly—but triggers federal environmental review under NEPA, which must occur before data center construction begins. Because DOD and U.S. Department of Energy (DOE) will not award new projects until mid-2025, DOD’s programmatic review is critical to reducing data centers’ NEPA requirements.²² The programmatic review can take place before sites or projects are selected and, if done appropriately, may substantially reduce the need for site-specific analysis after selection. This Administration’s energy EO (14154) has already taken steps to streamline NEPA, which might work in tandem with DOD’s programmatic review. However, EO 14154 does not eliminate the need for DOD’s review. The NEPA statute will require a threshold level of review in any event – and DOD programmatic review may satisfy this requirement more quickly than the alternative, which is for agencies to undertake lengthy new rulemaking to revise and update their NEPA procedures.

Option 2: Require additional review of categorical exclusions (CXs) for natural gas and oil

Under NEPA, federal agencies are required to draft detailed environmental impact statements (EISs) for all “major Federal actions” (generally including permit approvals) that are determined to “significantly affect the quality of the human environment”.²³ EISs can take 2 or more years for many projects. To determine whether an action would significantly affect the quality of the human environment, agencies must often separately prepare environmental assessments (EAs).

Pursuant to Section 109 of NEPA, agencies may establish categorical exclusions (CXs), which are categories of major actions that do not require any further environmental review (i.e., an environmental assessment (EA) or environmental impact statement (EIS)). To establish a categorical exclusion, the agency must determine that the relevant category of action typically does not have a significant effect on the human environment. EO 14141 directs agencies such as the U.S. Department of the Interior (DOI) to explore adding new categorical exclusions for geothermal energy and other forms of clean energy, as well as to leverage existing ones which could apply to AI infrastructure (e.g., transmission, substations).²⁴ It may be possible to unlock additional energy capacity by developing categorical exclusions for traditional energy sources (e.g., natural gas). This could streamline federal permitting processes and data center development timelines. Note, however, that establishing such exclusions could take significant time and further cost-benefit analysis is needed.

²² See generally this memorandum explaining programmatic NEPA review: Council on Environmental Quality, *Effective Use of Programmatic NEPA Reviews*, December 18, 2014.

²³ U.S. Environmental Protection Agency, “National Environmental Policy Act Review Process,” webpage, July 15, 2024; U.S. Department of Energy, “Categorical Exclusion (CX) Determinations,” webpage, undated.

²⁴ Executive Order 14141 §7(f)(iii), 2025.

Option 3: Activate Exemptions from the Endangered Species Act (ESA)

The ESA has historically created delays for infrastructure projects.²⁵ Section 7 of the ESA provides that federal agencies’ must insure that their actions are “not likely to jeopardize the continued existence of any endangered species or threatened species” or result in the destruction or adverse modification of “critical habitats” of such species.²⁶ Agency “action” is expansively interpreted to include federal projects, granting federal permits, licenses, contracts, rights-of-way, leases and promulgation of federal regulations.²⁷ Agencies must evaluate the effects of agency actions and consider alternatives to minimize those effects, as needed.²⁸ However, Section 7(j) of the ESA allows the DOD to exempt projects where the Secretary of Defense finds that such exemption is “necessary for ... national security.”²⁹ The DOD exemption can take effect only after the ESA’s required 90-day consultation process with the U.S. Fish and Wildlife Service – as such, it could be important to proceed with such consultations quickly for AI infrastructure on federal sites.³⁰

Option 4: Activate Clean Air Act (CAA) and Clean Water Act (CWA) Exemptions

The CAA and CWA impose several permit requirements on industrial projects, which can delay construction.³¹ The CWA requires section 404 permits to build near federal wetlands – “more significant impact” permits take about 217 days (7+ months) on average.³² The CAA mandates that new major sources of air pollution (or major modifications to existing sources) undergo New Source Review (NSR) permitting and install state-of-the-art emission controls before construction begins.³³ The NSR permitting process for large energy projects is “complex, time-consuming, and occasionally costly – the average processing time was about 420 days (well over a year) from application to permit issuance.”³⁴ These delays in obtaining air permits can postpone the start of construction and deter investments.³⁵

The CAA and CWA both give the President broad authority to exempt projects from those acts’ compliance requirements, so long as the President deems such exemptions to be “in the

²⁵ See, e.g., the Atlantic Coast Pipeline (ACP), a proposed 600-mile interstate natural gas pipeline, was halted after ESA litigation in 2018 (see Gilmer, 2018); Additionally, Nebraska Public Power District’s “R-Project”, a proposed 225-mile, 345-kV transmission line, was halted by courts pursuant to ESA litigation (see Doyle, 2024).

²⁶ Code of Federal Regulations, Title 16, Conservation; Chapter 35, Endangered Species; Section 1536, Interagency Cooperation.

²⁷ Code of Federal Regulations, Title 16, Chapter 35, Section 1536; Erin H Ward and Pervaze A Sheikh, *Endangered Species Act (ESA) Section 7 Consultation and Infrastructure Projects*, Congressional Research Service, R46867, August 4, 2021.

²⁸ Code of Federal Regulations, Title 16, Chapter 35, Section 1536; Ward, 2021.

²⁹ Code of Federal Regulations, Title 16, Chapter 35, Section 1536(j); note that this exemption has never been invoked, and it may not fully exempt agencies from Section 9’s prohibitions on “taking” endangered species.

³⁰ Further analysis its feasibility and the cost-benefits of a such a decision is needed; Benjamin Rubin, “Calling on the God Squad. Endangered Species Law and Policy,” webpage, April 13, 2024.

³¹ CAA, CWA, and ESA were cited by a survey of domestic manufacturers by the DOC as being among their most significant production bottlenecks. Bottlenecks to manufacturing are particularly relevant for industrial projects that rely on manufacturing inputs such as solar panels or AI chips. U.S. Department of Commerce, 2017.

³² Nicole T Carter, *The Army Corps of Engineers’ Nationwide Permits Program: Issues and Regulatory Developments*, Congressional Research Service, R97223, January 12, 2017.

³³ These may apply to data centers’ backup generators, construction work for transmission lines, and more.

³⁴ Arthur G Fraas, “EPA’s New Source Review Program: A Look at Permit Processing Time,” webpage, February 10, 2025.

³⁵ Fraas, 2025.

paramount interest” of the United States.³⁶ These exemptions are limited to actions by “agencies” and “federal instrumentalit[ies] of the executive branch.”³⁷ Although privately built data centers (even on federal lands) may not count as “federal instrumentalities,” a legal argument could be made that transmission projects benefiting from Section 1222 public/private partnerships (discussed below) would meet the bar. Further legal analysis is needed to assess this potential argument. The Executive Office of the President (EOP) could also explore whether projects involving significant DPA-initiated purchases may count as “federal instrumentalities.”³⁸

Option 5: Consider using Defense Production Act (DPA) authorities

DPA Title I prioritization and allocation

The President may be able to use DPA prioritization and allocation authority to accelerate AI infrastructure. Title I of the DPA also allows the President – as “necessary or appropriate” for national defense – to: (i) require companies to prioritize fulfilling U.S. government (USG) contracts; or (ii) allocate materials and services.³⁹ EOP could explore several possible uses of these authorities to expand energy supplies for AI data centers. First, the USG could leverage Title I authorities to require utilities to accelerate providing interconnections to the electric grid for energy resources needed to power AI data centers on federal sites, to the extent necessary for national security. A second potential option may be for the USG to require energy companies to reserve energy resources for AI data centers, ensuring that they receive power in periods of energy scarcity – as required for national security. For example, if natural gas supplies are constrained by a severe shortage, the President could potentially require gas pipelines to deliver fuel preferentially to power plants serving AI data centers to the extent necessary for national security.

DPA Title III notwithstanding clause

Title III of the DPA may permit the President to purchase materials needed for national defense notwithstanding federal and state laws that might otherwise constrain such purchases. Subsection 303(b) states that Presidents may make such purchases “without regard to the limitations of existing law.”⁴⁰ Similar “notwithstanding” provisions have been used to exempt large projects from environmental laws. For example, when President Trump declared a national emergency at the southern border, his administration used 10 U.S.C. §2808’s “notwithstanding” clause to avoid triggering NEPA-required procedures before constructing portions of a wall at the U.S. southern border.⁴¹ This DPA provision has not been used to exempt large-scale projects before, so further legal analysis is needed to determine under what circumstances the President may

³⁶ Code of Federal Regulations, Title 42, The Public Health and Welfare; Chapter 85, Air Pollution Prevention and Control; Subchapter I, Programs and Activities; Part A, Air Quality and Emission Limitations; Section 7418, Control of pollution from Federal facilities; Code of Federal Regulations, Title 33, Navigation and Navigable Waters; Chapter 26, Water Pollution Prevention and Control; Section 1323, Federal Facilities Pollution Control.

³⁷ Code of Federal Regulations, Title 42, Chapter 85, Section 7418; Code of Federal Regulations, Title 33, Chapter 26, Section 1323.

³⁸ Further analysis its feasibility and the cost-benefits of a such a decision is needed.

³⁹ Note that the President must make certain factual findings about the resources’ scarcity to activate these authorities; Alexandra G Neenan and Luke A Nicastro, *The Defense Production Act of 1950: History, Authorities, and Considerations for Congress*, Congressional Research Service, R43767 October 6, 2023.

⁴⁰ Neenan, 2023.

⁴¹ *Trump v. Sierra Club*, 140 S. Ct. 2620, 2020.

leverage this authority to avoid triggering or waive NEPA or other federal environmental requirements for AI infrastructure projects.

B. Grid Access and Building Transmission Lines

It is important to ensure that the federal sites have reliable energy transmission access. EO 14141 took some actions to this end, such as by requiring transmission providers to report locations with surplus capacity to connect new energy projects to the grid. Potential expansions to these policies include: (i) leveraging DOE’s Section 1222 authority to facilitate construction of certain electric power transmission facilities, especially where state permitting processes may significantly delay necessary transmission projects; and (ii) tasking DOE to engage utilities serving AI infrastructure to accelerate their interconnection processes.⁴²

Option 6: Avoid triggering state regulatory processes that delay transmission projects

Additional statutory authorities may be available to accelerate the siting and upgrading of transmission lines. EOP might consider commissioning a thorough legal review of statutes, potentially enabling the executive branch to avoid triggering state regulatory reviews that can delay transmission projects. One potential example is Section 1222 of the Energy Policy Act (EPA) of 2005, which enables DOE to partner with companies to accelerate transmission projects. Prior DOE legal analysis suggests that Section 1222 may permit the executive branch to:⁴³

Avoid triggering lengthy reviews by state utility commissions by leveraging public/private partnerships. Section 1222 of the EPA allows DOE to establish public/private partnerships for transmission projects and does not waive DOE’s sovereign immunity. Pursuing transmission projects via these public/private partnerships— for which sovereign immunity is not waived—thus avoids triggering state level public utility commission (PUC) reviews, which can last many years.⁴⁴

Activate backstop eminent domain authorities. Section 1222 allows the DOE to exercise eminent domain over private parcels of land needed for transmission development, as a backstop, if doing so serves a public purpose. These authorities would function similarly to eminent domain powers that state PUCs typically grant private developers for transmission projects.

Activating this authority requires finding that the transmission project is “necessary to accommodate ... [a] projected increase in demand for electric transmission capacity” – a condition that may be met by expected future AI infrastructure demand.⁴⁵ EOP could explore

⁴² Public Law 109-58, Energy Policy Act; Section 1222, Third-Party Finance, August 8, 2005.

⁴³ Further analysis its feasibility and the cost-benefits of a such a decision is needed. Section 402 of the American Recovery and Reinvestment Act and Section 40106 of the Infrastructure Investment and Jobs Act grant DOE similar authorities; U.S. Department of Energy, *Summary of Findings: Plains & Eastern Clean Line Project*, March 25, 2016.; Justin Gundlach, *Transmission Siting Reforms in the Infrastructure Investment and Jobs Act of 2021*, Institute for Policy Integrity, December 2021, p. 4, 9-10.

⁴⁴ DOE possesses independent authorities to perform—and in some cases does perform—“cost allocation” functions for transmission lines normally handled by state public utility commissions. See: Western Area Power Administration, “Rates and Repayment,” webpage, September 20, 2024.

⁴⁵ National Law Review, “DOE Participation In Plains & Eastern Clean Line Could Open Doors for Other Transmission Projects,” webpage, March 30, 2016.

these legal authorities further to determine whether they may permit the USG to establish a program for AI developers to apply and partner with DOE on transmission projects that serve frontier AI data centers.

Option 7: Task DOE to engage utilities on overhauling interconnection processes

The DOE could help expedite utility interconnection processes for AI infrastructure on federal sites. To provide sufficient energy to serve their data centers, AI developers may need to draw on power from multiple utilities, which will require significant interconnections.⁴⁶ However, utilities' interconnection processes often take 4-6 years.⁴⁷ Utility companies could take several steps to streamline interconnection for AI infrastructure — such as reforming queue processes, adopting faster grid resilience tests, or prioritizing AI infrastructure over other applicants. While only the Federal Energy Regulatory Commission, an independent regulator, can require utility companies to reform their processes, DOE can negotiate voluntary reforms with utilities serving federal sites. Potential reforms by utilities could include: (i) prioritizing generation resources that satisfy “resource adequacy” needs or bring national security benefits; (ii) using automated tools to adjudicate interconnection applications; and (iii) information disclosures that facilitate auctions of low-cost interconnection capacity.

C. Designating Federal Lands for AI Infrastructure

Infrastructure projects built on federal lands are not subject to many state and local-level regulations, such as permitting requirements. The USG has previously directed DOD and DOE to make sites federal lands available for AI data centers, streamlining permitting requirements for these projects.⁴⁸ This policy could be expanded by: (i) ensuring timely completion of DOD and DOE's federal site-selection process; (ii) and potentially removing clean power requirements. We discuss these potential options next.

Option 8: Ensure timely completion of DOD and DOE federal site-selection processes

The strategy laid out by EO 14141 and EO 14154 – making federal lands available for AI infrastructure – hinges on the DOD and DOE's ability to quickly identify suitable sites for AI data centers. DOD and DOE could screen for potential issues that may cause future delays and litigation (e.g., the need for substantial new transmission infrastructure or the presence of historical properties). Any delay in site selection could cascade through the entire development timeline, as most other work cannot begin until sites are chosen and leased.

Option 9: Remove requirements related to “clean power”

EO 14141's strict clean power requirements restricts the available options for bringing new energy sources online quickly. While the U.S. has abundant traditional energy sources – particularly natural gas – EO 14141 makes them challenging to use for AI infrastructure. Almost none of EO 14141's mechanisms for streamlining permitting may apply to natural gas, oil, or

⁴⁶ Specific statutes relevant are EPA act sec 1222, ARRA sec 402, and IIJA sec 40106; Srivathsan, Bhargs, *AI Power: Expanding Data Center Capacity To Meet Growing Demand*, McKinsey & Company, October 29, 2024.

⁴⁷ Paul Ciampoli, “DOE Offers Funding to Accelerate Interconnection Process Through Utilization of Artificial Intelligence,” webpage, November 26, 2024.

⁴⁸ See e.g., Section 4 of EO 14141, which directs the DOD and DOI to make at least six federal sites available for AI data centers in a one-off rental auction, streamlining the permitting process for these locations.

coal unless accompanied by highly costly carbon capture technology. EO 14141 requires all companies leasing federal sites to “procur[e] sufficient new clean power generation resources ... to meet [all of] the frontier AI data center’s planned electricity needs.” If such requirements were removed, more companies could potentially access a wider range of energy sources and land for power production.

D. National Security-Related Requirements for AI Infrastructure Developers

To ensure that the USG harnesses AI for national security, EO 14141 requires leasing entities to: (i) adhere to cybersecurity standards, which may be needed to prevent foreign theft or sabotage of AI infrastructure; (ii) allow the USG to procure AI models at or below market rates; and (iii) sign memorandum of understanding (MOU) with the DOC to collaborate on AI evaluations. These policies could be expanded by requiring leasing entities to report risk factors of the personnel they hire – including nationality and work history – to counter potential espionage. Although these are not directly energy policies, they could support and strengthen companies that play a key role in unlocking power for AI, e.g., by improving their security.

Option 10: Require companies leasing federal sites to report security risk factors of their personnel

As frontier AI capabilities improve, AI infrastructure may increasingly become a high-priority target for cyber operations by China, Russia, and other U.S. adversaries.⁴⁹ Leading AI companies are not on track to achieve the security required to protect against well-resourced actors.⁵⁰ If adversaries gain physical access to AI data centers – including those developed on federal sites – they may steal AI model weights or sabotage AI systems by inserting vulnerabilities.⁵¹ Mandatory reporting of personnel risk factors – including nationality and work history – could aid AI security efforts. RAND’s ongoing research into AI facility personnel security is available upon request to inform the development of these protocols.

E. Conclusion

This paper identifies potential policy options that may help reduce the U.S. AI energy gap. The energy policy landscape is complicated, with overlapping federal, state, and local equities. The costs and benefits of these changes, as well as their legal feasibility, should be fully assessed prior to implementation. This full assessment was beyond the scope of this working paper. We suggest further study on these policy options including their costs, benefits, and legal feasibility.

⁴⁹ Luke Barr, “China, Russia and Iran ‘Attempting To Illegally Acquire’ US Technology, Top DHS Official Says,” ABC News, 28 July 2023; Farivar, Masood, “FBI Warns About China Theft of US AI Technology,” *Voice of America*, 28 July 2023.

⁵⁰ See Sella Nevo, Dan Lahav, Ajay Karpur, Yogev Bar-On, Henry Alexander Bradley, and Jeff Alstott, *Securing AI Model Weights: Preventing Theft and Misuse of Frontier Models*, RAND Corporation, RR-A2849-1, 2024.

⁵¹ Nevo et al, 2024.

Abbreviations

AI	artificial intelligence
CAA	Clean Air Act
CX	Categorical Exclusion
CWA	Clean Water Act
DOC	Department of Commerce
DOE	Department of Energy
DOI	Department of the Interior
DOD	Department of Defense
DPA	Defense Production Act
EA	Environmental Assessment
EIS	Environmental Impact Statement
EO	Executive Order
EOP	Executive Office of the President
ESA	Endangered Species Act
GW	Gigawatt
NEPA	National Environmental Policy Act
U.S.	United States
USG	U.S. government

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